

Project 10: Fine-Tuning Studio for Domain Adaptation

Instruction Data Validation & Parameter-Efficient LoRA Tuning

1. Executive Overview

The Fine-Tuning Studio manages parameter-efficient fine-tuning (PEFT) workflows using Low-Rank Adaptation (LoRA). It validates domain instruction JSONL data, configures rank ($r=8$) and scaling factor ($\alpha=16$), tracks epoch training loss, and evaluates base vs. fine-tuned model performance gains (+44.9% accuracy gain).

2. System Architecture & Workflow Pipeline

- Dataset Validator & Tokenizer Inspector (domain_data.jsonl)
- LoRA Configurer ($r=8$, $\alpha=16$, target_modules=[q_proj, v_proj])
- LoRA Adapter Trainer (Multi-Epoch Loss Tracking: 2.45 -> 0.54)
- Base vs. Adapted Model Evaluator (+44.9% Domain Accuracy Gain)

3. API Endpoints Specification

Method	Endpoint	Description
GET	/	Serves Fine-Tuning Studio web UI
GET/POST	/api/run-fine-tune	Executes LoRA adapter training and returns history & evaluation metrics

4. Deployment & Automated Verification

Standalone Uvicorn Launch:

```
cd 10_fine_tuning/src && python -m uvicorn server:app --host 127.0.0.1 --port 8010
```

Automated Test Suite Execution:

```
python -m pytest 10_fine_tuning/tests/
```